

Hugo Blanc

PhD Candidate in Computer Vision and 3D Reconstruction

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GitHub: github.com/hugobl1
Location: Paris, France



Research interests

Physically-based rendering; novel view synthesis; 3D surface reconstruction from images and point clouds; multi-view geometry; geometry processing.

Education

Mines Paris – CAOR

PHD CANDIDATE — 3D COMPUTER VISION AND RENDERING

Supervisors: Jean-Emmanuel Deschaud, Alexis Paljic

2022 – 2026

Paris, France

- Research on 3D scene reconstruction and differentiable volumetric rendering using local Gaussian/RBF representations.
- Developed RayGauss (WACV 2025, Best Paper – Algorithms) and RayGaussX (ICCV 2025), Gaussian-based ray casting methods for differentiable rendering that improve both quality and efficiency over Neural Radiance Fields, while producing higher-quality renderings with fewer artifacts than 3D Gaussian Splatting.
- Developed a global implicit surface model from local RBFs for accurate and BVH-accelerated geometry reconstruction, resulting in the publication of “Hermite Radial Basis Function for Surface Reconstruction via Differentiable Rendering” (CVPR 2026, Highlight).

ENS Paris-Saclay

MASTER’S DEGREE — MVA (MATHEMATICS, VISION, LEARNING)

2021 – 2022

Paris, France

- High-level M2 Master’s programme providing courses related to the acquisition and processing of digital data and their automatic interpretation.
- Selected coursework: Computational Statistics, Object Recognition and Computer Vision, Convex Optimization, Point Clouds and 3D Modeling (NPM3D), Introduction to Digital Imaging, Kernel Methods for Machine Learning, Graphs in Machine Learning, Deep Learning.

ENSTA Paris

DIPLÔME D’INGÉNIEUR (MSC EQUIVALENT)

2019 – 2022

Palaiseau, France

- Applied Mathematics track — ranked in the top 5% of the cohort.

Lycée Claude Fauriel

CLASSE PRÉPARATOIRE MPSI – MP* (MATHEMATICS AND PHYSICS)

2016 – 2018

Saint-Etienne, France

- Intensive undergraduate program in mathematics and physics.

Research and professional experience

Mines Paris – CAOR

PHD CANDIDATE

2022 – 2026

Paris, France

- Main research contributions include RayGauss (volumetric Gaussian-based ray marching), RayGaussX (accelerated Gaussian-based ray marching), and a global implicit surface model from local RBFs (see Selected publications).
- Daily development in Unix/Linux with C++, Python, CUDA and OptiX, with extensive use of PyTorch, OpenCV and Git.

Mines Paris – CAOR

RESEARCH INTERN

Supervisor: Jean-Emmanuel Deschaud

Apr. 2022 – Sep. 2022

Paris, France

- Investigated approaches for accelerating NeRF-style volumetric rendering for interactive novel view synthesis, leveraging 3D point cloud data acquired from LiDAR sensors or Multi-View Stereo.

University of Cambridge

RESEARCH ASSISTANT

Supervisor: Angelica Aviles-Rivero

May 2021 – Aug. 2021

Cambridge, UK

- Developed methods for semi-supervised hyperspectral image classification with limited labeled data using Graph Neural Networks.

Forum Trium

DEVELOPMENT COORDINATOR

Dec. 2019 – Mar. 2021

Palaiseau, France

- Organised a company career forum for engineering students, including contacting and recruiting partner companies and coordinating event logistics.

Selected publications

- **RayGauss: Volumetric Gaussian-Based Ray Casting for Photorealistic Novel View Synthesis**, WACV 2025 (Best Paper Award). Project page: raygauss.github.io.
- **RayGaussX: Accelerating Gaussian-Based Ray Marching for Real-Time and High-Quality Novel View Synthesis**, ICCV 2025. Project page: raygaussx.github.io.
- **Hermite Radial Basis Function for Surface Reconstruction via Differentiable Rendering**, CVPR 2026 (Highlight).

Teaching and supervision

- Taught the Master's-level course *Apprentissage Artificiel* (machine learning and deep learning) at Mines Paris — course page: people.minesparis.psl.eu/~fabien.moutarde/ES_MachineLearning.
- Taught the MVA course *Nuages de Points et Modélisation 3D (NPM3D)* at ENS Paris-Saclay, on processing and using 3D point cloud data — course page: master-mva.com/cours/nuages-de-points-et-modelisation-3d.
- Supervised a Master's intern on novel view synthesis for virtual reality exploration of reconstructed environments.

Talks and outreach

- Talks and posters at international conferences such as WACV and ICCV.

- Presentation at GT Rendu 2025, a yearly rendering workshop organised by Inria Paris — information: gtrendu.blogspot.com.
- Public demonstrations of research during the European Heritage Days at Mines Paris.

Skills

Programming and tools: C++, Python, CUDA, OptiX, PyTorch, OpenCV, git, Optuna, TensorBoard.

Research topics: 3D reconstruction, implicit neural rendering, Gaussian-based and RBF-based representations, differentiable volumetric rendering, multi-view geometry, machine learning.

Languages: French (native), English (fluent), Spanish (working knowledge), German (basic).